Sterling Fitness — Technique Tables (FIT-T)

Sterling Fitness · SterlingLife · draft v0.2 · 2026-10-27 · for owner review · Method only: no member results in this file. Generated from fitness_spec.py.

Terms

Term	Code name	Meaning
Body part	`body_part`	What gets scored, e.g. Legs.
Movement	`movement`	The kind of motion, e.g. Knee bend.
Main muscles	`muscles`	The muscles a movement works. Described, never scored.
Exercise	`exercise_id`	What the member does and logs, e.g. Leg Press (legPress). The id never changes; the name may.
Set	`sets`	One round of an exercise, as logged in the app.
Age band	`age_band`	The member's age range. It picks targets and tests.
Target	`target`	The weekly sets a body part should get. Same for everyone in an age band. Every program must meet it.
Limit	`limit`	The most weekly sets a program may assign a body part. Same for everyone in an age band.
Program	`program`	The exercises and weekly sets assigned to one member for a period. Built by code (N3), approved by the member (G1). Each version starts on a Sunday.
Program score	`program_score`	How much of the week's program the member did. The member's strength score.

Code key

Code	Meaning
N0-N3	Intake: member record, age-band planner, evidence store, program builder
ST1-ST5 ...	Agents, by lane (ST = Strength, CA = Cardio, ...)
R-xx	Results set for one lane
X0-X3	Checks: sources, contradictions, scoring, re-tests
S1-S4	Outputs: 925 score, assessment, wiki, plan
G1-G4	Approval gates: a person must approve before work continues
D# / O#	Deviation (code differs from this table) / open owner decision

Exercise model (proposed)

Body part → Movement → Exercise. Every exercise belongs to one movement. Every movement belongs to one body part. Muscles are described, never scored. No row has a published source yet. Unmarked exercises are not in the app yet.

Upper body

Movement	Main muscles	Exercises: name, id
Push forward (push)	Chest, front shoulders, triceps	Chest Press `chestPress` · **in app** Push-up `pushUp` · **logged, skipped (D4)** Wall Push-up `wallPushUp`
Push overhead (push)	Shoulders, triceps	Overhead Press `military` · **in app** — App name today: Military Seated Dumbbell Shoulder Press `seatedDumbbellPress`
Arm straighten (push)	Triceps	Triceps Extension `triceps` · **in app** — App name today: Triceps Triceps Pushdown `tricepsPushdown`
Pull down (pull)	Upper back (lats), biceps	Lat Pulldown `lat` · **in app** — App name today: Lat Overhead Pull `overheadPull` · **in app** — Works the lats (O3) Assisted Pull-up `assistedPullUp`
Pull back (pull)	Middle back, rear shoulders, biceps	Seated Row `row` · **in app** — App name today: Row Band Row `bandRow`
Arm bend (pull)	Biceps	Biceps Curl `bicepCurls` · **in app** Hammer Curl `hammerCurl`
Shoulder lift (pull)	Upper back (upper trapezius)	Shoulder Crunch `crunch` · **in app** — App name today: Crunch. Not a stomach crunch. O3

Core

Movement	Main muscles	Exercises: name, id
Trunk curl	Front stomach muscles (rectus abdominis)	Ab Crunch `abs` · **in app** — App name today: Abs Ab Machine Crunch `abMachineCrunch`
Trunk twist and side bend	Side stomach muscles (obliques)	Side Bend `sideBend` Standing Cable Twist `standingCableTwist`
Back raise	Lower back	Bird Dog `birdDog` Back Extension `backExtension`
Hold steady	Deep stomach muscles, lower back	Plank `plank` Wall Plank `wallPlank`

Legs

Movement	Main muscles	Exercises: name, id
Knee bend	Quadriceps, glutes	Leg Press `legPress` · **in app** Squat `squat` · **logged, skipped (D4)** Sit-to-Stand `sitToStand`

Hip bend	Glutes, hamstrings	Hip Bridge `hipBridge` Hip Hinge `hipHinge` Deadlift `deadlift`
Knee curl	Hamstrings	Hamstring Curl `hamstringCurl`
Leg out to the side	Hip abductors (glute medius, tensor fasciae latae)	Side Leg Raise `sideLegRaise` Hip Abduction Machine `hipAbductionMachine`
Leg in	Adductors (inner thigh)	Hip Adduction Machine `hipAdductionMachine` Ball Squeeze `ballSqueeze`
Knee lift	Hip flexors	Standing Knee Raise `standingKneeRaise` March in Place `marchInPlace`
Heel raise	Calves	Heel Raise `heelRaise`
Toe raise	Shin (tibialis anterior)	Toe Raise `toeRaise`

Rules

1. Log by exercise. Score the program by exercise. Check and report targets by body part.
2. Each exercise has a permanent id. Never match logs by name.
3. Compare weight and progress only within the same exercise (ST4).
4. Code builds the program from written rules (N3). A language model never does.
5. The member approves the program before it starts (G1).
6. Score against the program. Every program meets the targets, so a member who falls short of the program scores lower.
7. Only the assigned exercise counts toward the program score. A substitute is logged and counts toward the target report, but earns no program score.
8. A program change starts on a Sunday. Each week is scored against the program version in force that week, and the score stores that version.
9. Every movement and exercise needs a published source before it ships.

ST1 · Training Volume & Muscle Balance (ages 50-64 · 65+)

Scores how much of the week's program the member did, and reports whether each body part got enough strength work, often enough, for the member's age band. One agent for all ages: only the target changes, and N1 supplies the band.

Step	What it does	Rule or source	Status
ST1.1 Load program and target	Loads the program version in force this week (approved at G1, started on a Sunday), and the targets for the age band from N1. Stops and reports if there is no approved program or target.	NSCA2019, BORDE2015	**proposed**
ST1.2 Find body part	Looks up the body part of each logged exercise in the exercise model.	Program rule	**built**
ST1.3 Count sets	Adds up sets per exercise and per body part, Sunday to Saturday.	Program rule	**built**

ST1.4 Count days	Counts the days each body part was trained. Rule: 2 or more.	WHO2020, PAG2018	**proposed**
ST1.5 Program score	Per assigned exercise: sets done of that exercise / sets assigned, capped at 1. Program score = 100 x the average over the assigned exercises. The member's strength score.	Program rule	**proposed**
ST1.6 Target report	Per body part: sets / target, capped at 1 (upper body: push and pull each). Reported, not scored.	NSCA2019, BORDE2015	**built**
ST1.7 Report	Sends the program score, the program version, sets, days, age band, the target report and the target's sources to R-ST.	Program rule	**proposed**

Deviations: where the code differs

Id	Agent	Detail
D1	ST1	The fitness_analytics.py docstring describes a 'hard set' rule (3 or fewer reps in reserve, or at least 60% of estimated 1RM). The code counts every logged set, because the app records no weight or effort. Fix: reword the docstring to match the code. Weight and effort belong to ST4.
D2	ST1	Training days per body part (ST1.4) are not counted or scored.
D3	ST1	The weekly targets in the code are one placeholder table for every age (legs 10, push 8, pull 8, core 5), with no source. Decided: 6 a week per body part, push and pull 6 each, limit 9 (O1, O2, O4).
D4	ST1	Squats and push-ups are logged but skipped by the code (EXCLUDED_EXERCISES). The exercise model counts them.
D5	ST1	The scoring code uses four groups (lower_body, upper_push, upper_pull, core) and averages four. The model has three body parts (legs, upper_body, core).
D6	ST1	The scoring code matches logs by the app's names (e.g. 'Military'), not by id. Switch matching to ids before any app name changes, or logged sets stop counting.
D7	ST1	The scoring code scores sets against targets (ST1.6). The method scores the program (ST1.5). Until the code changes, the strength score is the target score, and no program is stored.

Owner decisions (not build instructions)

Id	Agent	Status	Decision
O1	ST1	decided 2026-10-27	Ages 50-64: no study on weekly sets was found for this age group. Decided: use the 65+ range (4-9) and label it 'not validated under 65'. It is the lower, safer dose.
O2	ST1	decided 2026-10-27	The weekly target. Decided: 6 sets per body part, e.g. 3 sets on 2 days or 2 sets on 3 days. Limit: 9, the top of the range.
O3	ST1	decided 2026-10-27	Which movement are Overhead Pull and Crunch (shoulder crunch)? Decided: both are upper body, pull. Overhead Pull works the lats (Pull down). Shoulder Crunch works the upper back, upper trapezius (Shoulder lift).
O4	ST1	decided 2026-10-27	Upper body holds both push and pull. Decided: the upper body target counts push and pull separately (6 sets each), using each movement's side.

O5	ST1	decided 2026-10-27	Does following the program count toward the score? Decided: yes, it is the only score (ST1.5), and future points build on it. Assigned exercises are program components; a substitute earns nothing. N3 builds every program at or above target, so falling short of the program scores lower. Program changes start on Sunday. The program is weekly sets per exercise, with each body part on 2 or more days.
O6	ST1	open	A body part trained on only one day in a week. Recommended: its exercises earn at most half their program credit that week, so the 2-day rule counts without a separate score.

Appendix A: step notes

Step	Note
ST1.1	First, because the band is known before ST1 runs. Range: 2-3 sets per body part per session, 2-3 sessions a week, so 4-9 sets a week. Target 6 sets a week per body part; upper body counts push 6 and pull 6. Limit 9. Ages 50-64 use the 65+ range, labelled 'not validated under 65'. The code still uses a placeholder table. See D3, O1, O2, O4.
ST1.2	The guidelines (WHO2020, PAG2018) ask for all major muscle groups; the three body parts are the program's own split. See D4, D5.
ST1.3	Every logged set counts, substitutes included. See D1.
ST1.4	Same rule for every age band. How a body part trained on one day affects the program score: see O6. Not counted today: D2.
ST1.5	Matched by exercise id, so a substitute earns nothing here. The cap stops extra sets of one exercise from making up for one that was skipped, and gives no reason to overtrain. Sets only: the app records no weight or effort (ST4). See O5, D7.
ST1.6	Built today as the score itself (D7). Shows whether the body got enough work whatever the program said.
ST1.7	X0 needs the sources to check every result. The program version keeps old scores fixed when the program changes.

Appendix B: sources

Id	Citation	Ages covered	Used for	How far checked
WHO2020	Bull FC, Al-Ansari SS, Biddle S, et al. World Health Organization 2020 guidelines on physical activity and sedentary behaviour. Br J Sports Med 2020;54(24):1451-1462. doi:10.1136/bjsports-2020-102955	Adults 18-64 and older adults 65+, separate recommendations	Muscle-strengthening work on all major muscle groups, 2 or more days a week	Recommendation known; re-read the text before approval
PAG2018	Piercy KL, Troiano RP, Ballard RM, et al. The Physical Activity Guidelines for Americans. JAMA 2018;320(19):2020-2028. doi:10.1001/jama.2018.14854	US adults and older adults	Same 2-days-a-week, all-major-muscle-groups rule as WHO2020	Recommendation known; re-read the text before approval
NSCA2019	Fragala MS, Cadore EL, Dorgo S, et al. Resistance training for older adults: position statement from the National Strength and Conditioning Association. J Strength Cond Res 2019;33(8):2019-2052. doi:10.1519/JSC.0000000000003230	Older adults (the NSCA's first statement for this group)	2-3 sets per major muscle group, 2-3 sessions a week	Citation confirmed (PubMed 31343601). Set and session figures confirmed only through a published summary; read the full text before approval

BORDE2015	Borde R, Hortobágyi T, Granacher U. Dose-response relationships of resistance training in healthy old adults: a systematic review and meta-analysis. Sports Med 2015;45(12):1693-1720. doi:10.1007/s40279-015-0385-9	25 randomized trials, mean age 65 and over	2-3 sets per exercise; 3 sessions a week	Citation and inclusion rule confirmed. Dose figures from secondary sources only, and they disagree on intensity (50-70% vs 70-79% of 1RM), so intensity is not used. Read the full text (PMC4656698) before approval
SCHOEN2017	Schoenfeld BJ, Ogborn D, Krieger JW. Dose-response relationship between weekly resistance training volume and increases in muscle mass: a systematic review and meta-analysis. J Sports Sci 2017;35(11):1073-1082. doi:10.1080/02640414.2016.1210197	Mostly young adults: NOT age-appropriate under the evidence standard	Context only: sets per muscle group per week is the standard volume unit; more weekly sets gave more growth up to 10+	Findings confirmed through the authors' published comments
RALSTON2017	Ralston GW, Kilgore L, Wyatt FB, Baker JS. The effect of weekly set volume on strength gain: a meta-analysis. Sports Med 2017;47(12):2585-2601. doi:10.1007/s40279-017-0762-7	Mostly young adults: NOT age-appropriate under the evidence standard	Context only: graded dose-response between weekly sets and strength	Citation and finding confirmed (PubMed 28755103)